TSE_DZHYNSA_BOT

Lab 2: data pipelines, versioning, baseline and MLflow

KSE MLOps · detecting hidden paid posts («джинса») in Ukrainian Telegram channels

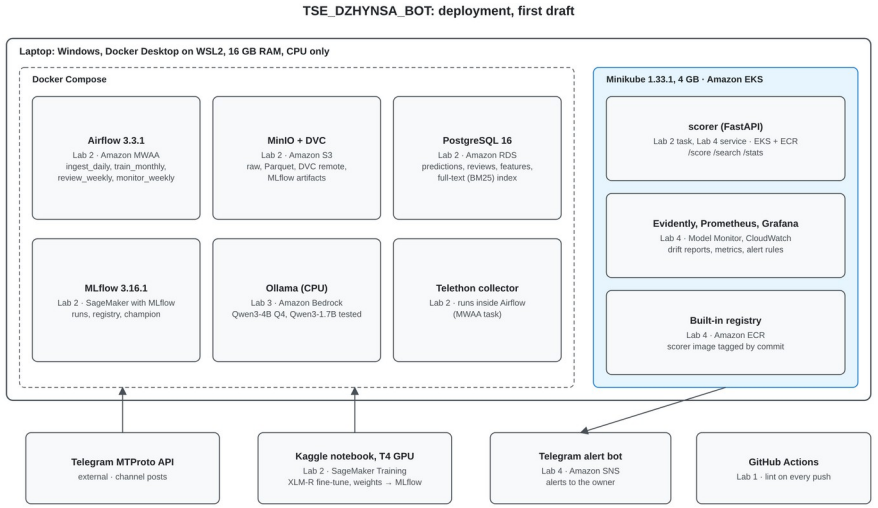
DP · November 2026

What Lab 2 delivers, and where the proof is

Requirement	Artifact in the repo	Proof
Raw data versioned, remote	data/raw.dvc · MinIO s3://dvc	make pull → md5 7cb35352…
ETL pipeline in Airflow 3	dags/ingest_daily.py (collect from Telegram)	daily, fully automated
Batch = logical date	CronDataIntervalTimetable(@daily, Kyiv)	run for 2026-09-10 → only that day
Idempotent rerun	delete-then-insert (warehouse.py)	760 / 760 rows
Quality gate fails on bad data	configs/quality.yaml · tests/fixtures/bad_batch	red quality_check
Feature table + serving source	features.py · table features (v1)	design.md 5.3
Rule vs model, time split	rule.py · train.py	PR-AUC 0.005 vs 0.749
Tracking + registry with lineage	MLflow (Postgres + MinIO)	dzhynsa-tfidf v1 @champion
Reproduce in one command	reproduce.py · make reproduce	diff 0.0000 ≤ 0.01
Tests in CI	tests/ (12) · lint.yml	ruff + pytest green

Stack: one docker compose

Service	Role	Port
PostgreSQL 16	warehouse · airflow · mlflow DBs	5432
MinIO (pgsty/silo)	DVC remote + MLflow artifacts	9000/9001
MLflow 3.16.1	tracking server + model registry	5000
Airflow 3.3.1	DAG ingest_daily (LocalExecutor)	8080



Deployment diagram (docs/diagrams)

- **One command:** docker compose up -d, then make pull / make train
- **Runs on a CPU-only laptop:** ~5 GB free disk, data is 21 MB
- **Secrets:** .env and .dvc/config.local, never in git; .env.example committed

Raw data under DVC

113,761
posts, 2026-06-01 … 10-27

85,367
with text

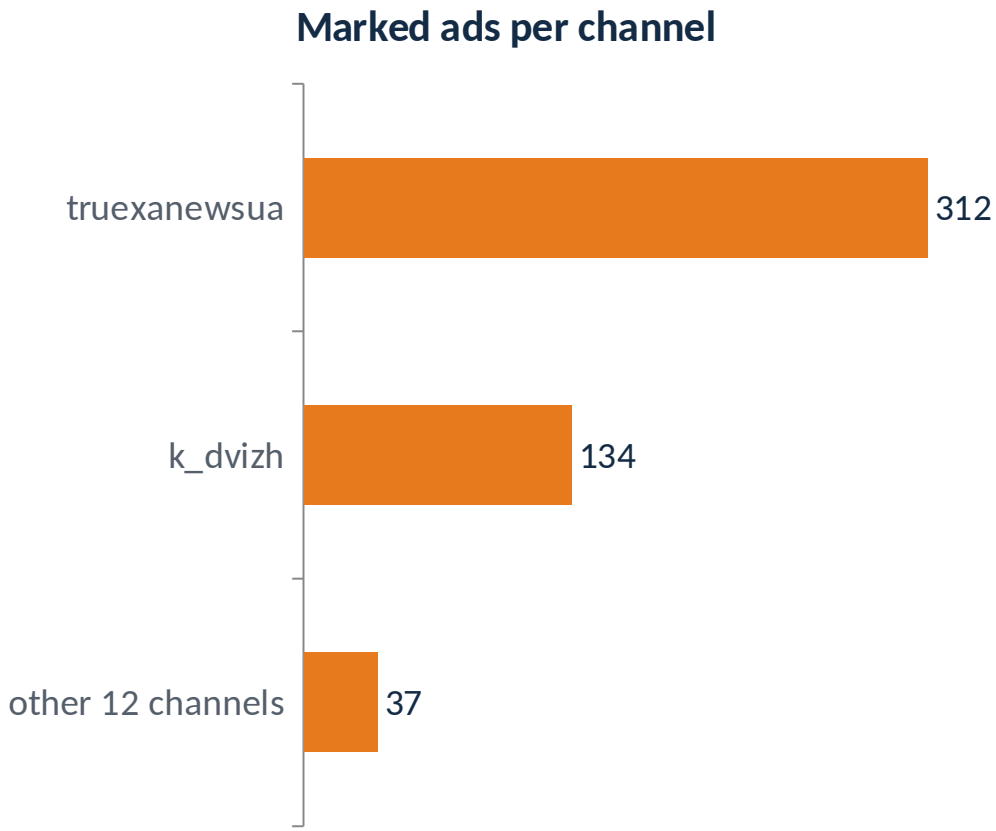
483
channel-marked ads

17.8 MB
2,086 files, no sampling

Item	Decision
Batch grain	One Kyiv day per channel: data/raw/<channel>/<YYYY-MM-DD>.jsonl.gz, sorted by id → same day = same bytes
Remote	S3 on MinIO (s3://dvc); config in .dvc/config (committed), keys in .dvc/config.local (not committed)
Clean clone	make pull (= dvc pull) → data/raw.dvc md5 7cb3535292c351068d6c6da584d5a45b.dir
Data version	That md5 is logged as data_version on every MLflow run and on the registered model
Label delay	ad: immediate (channel's own mark) · dzhynsa: ~1 week (owner review)

Labels: the channel's own ad mark

- **configs/ad_marks.yaml:** regexes for #реклама, #промо, #ad, «на правах реклами», «партнерський матеріал»
- **ad_marks.has_mark → label_ad = 1,** strip_mark cuts the mark out of the text (no leakage)
- **Cleaning rule fixed:** footer «… | Реклама» was read as a mark; it flagged 74 % of kyivoperat posts → regex '\bреклама\s*$' removed
- **Imbalance:** 483 ads of 85k texts (0.6 %) → PR-AUC, not accuracy; class_weight=balanced



Two channels carry 92 % of the labels: a known risk, channels to be replaced before Lab 3

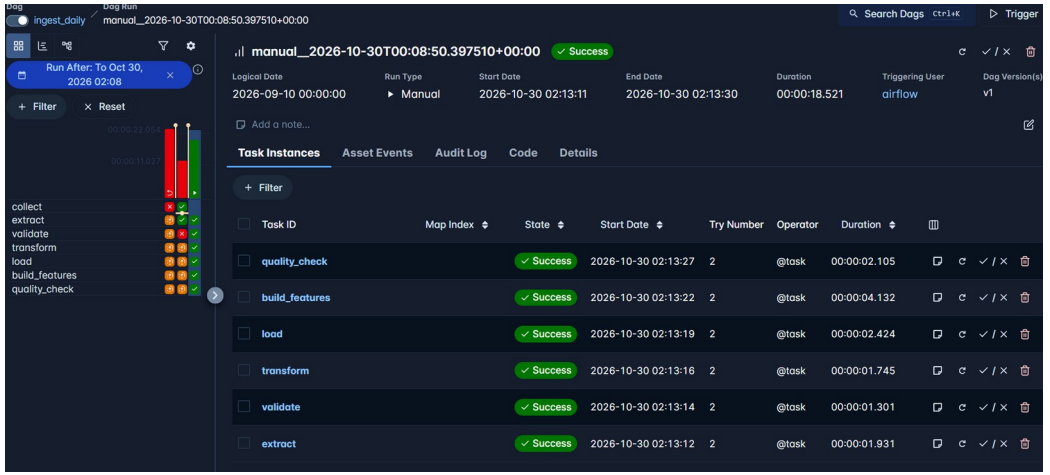
DAG ingest_daily: fully automated daily pipeline



Task	What it does
collect	Telethon download of the Kyiv day into data/raw (skipped without TG keys → DVC files)
extract · validate	read the day's files · schema check, batch not empty
transform	drop empty/duplicate texts, label_ad, strip the mark
load · build_features	delete the day, then insert; feature_version = v1
quality_check	row counts + thresholds; fails the run

Batch = logical_date in Europe/Kyiv, never now(); CronDataIntervalTimetable("@daily", Kyiv)

Backfill 2026-10-28 ... 11-09 ran through this DAG



Airflow: first try red (lock), rerun green end to end

Solved: "database is locked" in the Telethon SQLite session

Fix: temp copy of the session per run · max_active_tis_per_dag=1 · 2 retries

Idempotency: same interval twice, same rows

760

rows after run 1 (2026-09-10)

760

rows after run 2 (make rerun)

Run	posts_clean	features
first	760	760
second	760	760

warehouse.replace_day() — one transaction

```
with engine.begin() as con:
    con.execute("DELETE FROM {table} WHERE day = :day")
    df.to_sql(table, con, if_exists="append")
```

Collection is idempotent too: a day's file is rewritten whole, sorted by id, gzip without timestamp → identical bytes and the same DVC md5.

Quality gate fails for real

Check (configs/quality.yaml)	Threshold
Clean posts in the day	≥ 50
Channels with posts	≥ 5
Raw posts without text	≤ 70 %
Marked ads among clean posts	≤ 50 %
Rows: clean = loaded = features	equal

Bad batch, on purpose

tests/fixtures/bad_batch: 6 channels, 2026-09-11, 60 posts, all without text

Run with raw_dir = the fixture

→ **quality_check red:** “quality gate failed: clean rows < 50; empty-text share 1.00 > 0.7 …”

Screenshots: docs/screenshots/failed.png (red task) and success.png (green graph)

Why these checks: a day normally has ~600 posts from 14 channels; an empty or half-empty day means the collector or the source broke

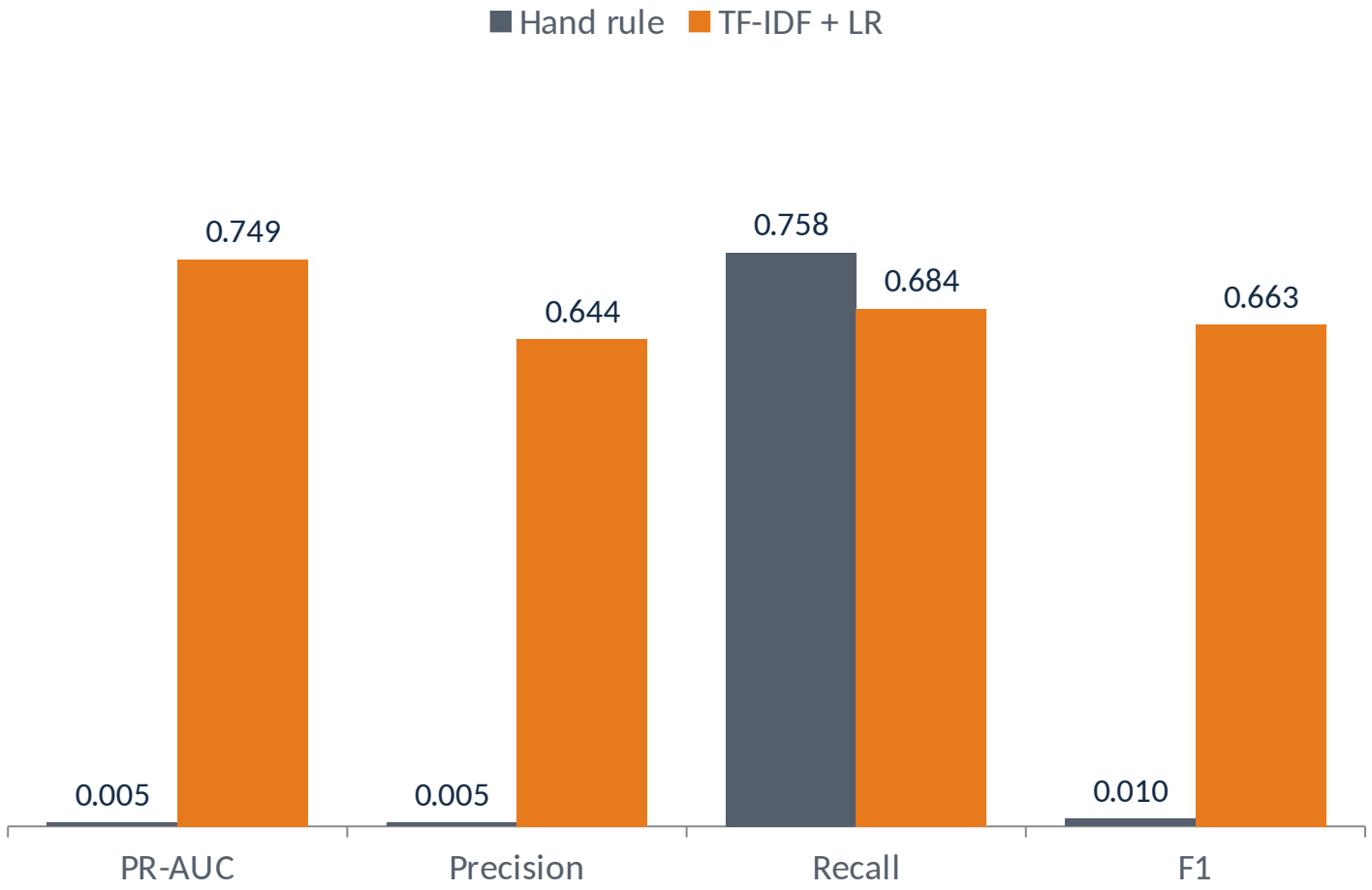
The ad-share check catches a broken mark regex (the footer bug would have tripped it)

Feature table and serving-time source

Feature	Training source	Serving-time source
text_clean (mark cut)	DVC raw → transform	the post itself (request)
n_links, has_invite_link, has_promo_code, has_phone	same code on the raw post	the post itself
text_len, is_forward, hour_of_day	same code on the raw post	the post itself
channel_ad_rate_30d	rolling 30 days before the post's day, today excluded	warehouse lookup in features (pipeline aggregate)

- **One code path:** features.py is imported by both train.py and the DAG → no training-serving skew by construction
- **Point-in-time:** channel_ad_rate_30d never sees the post's own day or later; covered by a pytest
- **Table features:** one row per post, feature_version = v1; Feast skipped (optional)

Baseline vs model on a time split



Split	Period
Train	Jun – Aug
Validation	Sep (threshold)
Test	Oct 1 – 27

Model wins: PR-AUC 0.749 vs 0.005

Threshold picked on validation for precision ≥ 0.70

Not a random split: the same ad is reposted for days → it would land in train and test

Test set: Oct 1–27; both runs logged to MLflow experiment dzhynsa-lab2

MLflow: tracking, registry, lineage

	Value
Backend store	PostgreSQL (db mlflow)
Artifact store	MinIO s3://mlflow
Experiment	dzhynsa-lab2: rule run + TF-IDF run
Logged per run	params, PR-AUC/P/R/F1, data_version, git_commit
Registered	dzhynsa-tfidf v1, alias @champion
Run id	cb943c6c63924ddcb6fce4ae2f2c1066

Lineage = two tags

data_version = DVC md5 of data/raw

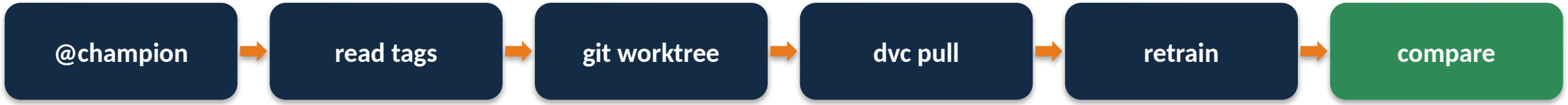
git_commit = code that trained it

On every run and on the registered version; Lab 4 loads models:/dzhynsa-tfidf@champion

Fix during the lab: logging the model with skops hung > 12 min → serialization_format="cloudpickle" (train.py, commit 373da4c).

Client env: MLFLOW_ENABLE_PROXY_MULTIPART_DOWNLOAD=false so load_model goes through the server, not minio:9000.

Reproduce the champion in one command



0.0000

PR-AUC diff (registered vs reproduced)

± 0.01

tolerance stated in README

1

command: make reproduce

```
uv run --env-file .env python -m tse_dzhynsa_bot.reproduce
```

Exit code 0 when |diff| ≤ 0.01, 1 otherwise → the reviewer sees pass/fail, not a number to eyeball.

Works from a clean clone: nothing depends on laptop paths; data comes from the MinIO remote.

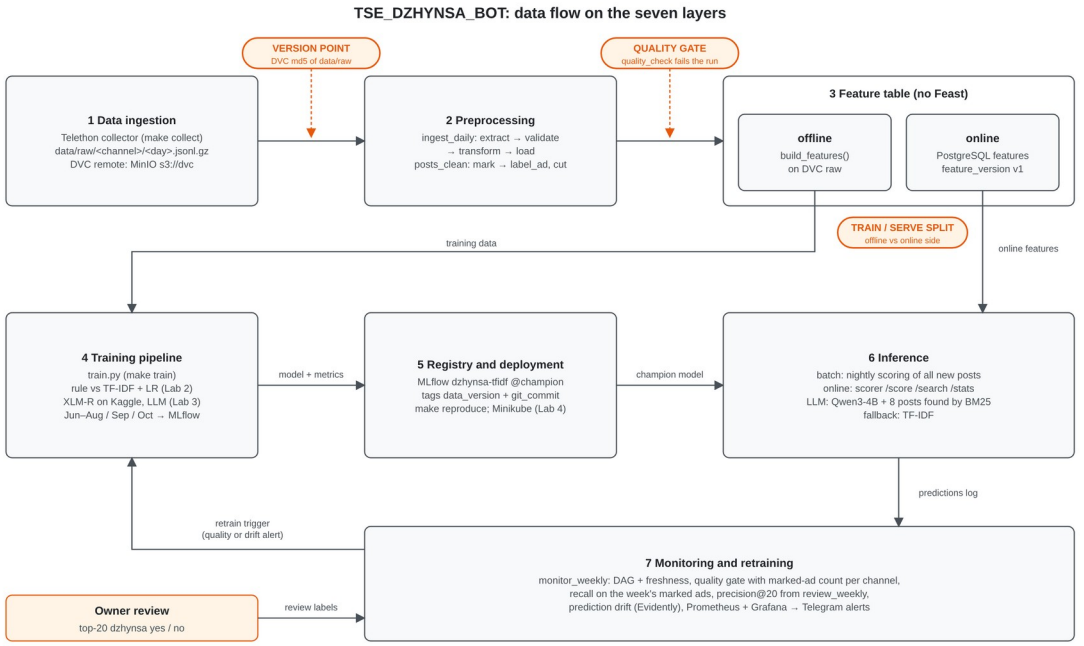
Tests, CI and the design doc

Test file	Covers	#
test_ad_marks.py	marks found and cut, footer is not a mark	4
test_features.py	validate, clean + dedup, columns, point-in-time	5
test_train.py	time split + training, threshold	2
test_collect.py	write_day is idempotent	1

CI (.github/workflows/lint.yml): ruff check + pytest on every push

design.md revised: 5.2 data · 5.3 techniques · 5.4 validation · 6.2 infra · 6.4 security · 6.5 privacy · 6.8 integration · 7.2 results

README: Lab 2 scorecard rows, pull command + checksum, 760/760, metrics



Data flow (design.md 6.1) with real component names

Lab 2 in one line

Data are versioned, the pipeline is automatic and safe to rerun, and every model traces back to its data and code

Done

DVC on MinIO, checksum in README

ingest_daily: collect → … → quality_check, fully automated

760 / 760 on rerun · red gate on bad batch

rule 0.005 vs TF-IDF 0.749 PR-AUC

dzhynsa-tfidf v1 @champion · reproduce diff 0.0000

Next: Lab 3

Replace channels that never mark ads

Fine-tuned XLM-R on Kaggle T4 → MLflow

LLM baseline: Qwen3-4B + BM25 examples

Eval set and ~50-post dzhynsa seed set